

Program : Diploma in Chemical Engineering	
Course Code : 4077	Course Title: Process Engineering Drawing
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To enable the students to interpret the process engineering drawings commonly used in chemical industries.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Engineering Graphics		Engineering Graphics	1
Computer fundamentals		Introduction to IT systems	1
AutoCAD		CAD Lab	2
Chemical equipments		Chemical Equipment Drawing	3

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret block flow diagrams	6	Applying
CO2	Interpret process flow diagrams	12	Applying
CO3	Interpret Piping and Instrumentation diagrams	21	Applying
CO4	Interpret piping plan and piping isometric drawings	6	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2			3				
CO3			3				
CO4			2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Interpret block flow diagrams		
M1.01	Sketch the overall block flow diagram of utility unit in a process plant including boiler house, cooling water unit, process air, instrument air unit, DM water unit and nitrogen generation unit.	3	Applying
M1.02	Sketch the block flow diagram of a boiler unit. Sketch the block flow diagram of a nitrogen generation unit.	3	Applying
CO2	Interpret process flow diagrams		
M2.01	Sketch Process Flow Diagram symbols for chemical equipments.	3	Applying
M2.02	Sketch the Process Flow Diagram of a boiler unit.	3	Applying
M2.03	Sketch the Process Flow Diagram of a nitrogen generation unit.	3	Applying
M2.04	Open ended experiment	3	Applying
	Series Test – I		
CO3	Interpret Piping and Instrumentation diagrams		
M3.01	Sketch common P&ID symbols for valves and pipe fittings as per ISA-5.1	3	Applying
M3.02	Sketch common P&ID symbols for lines as per ISA-5.1.	1	Applying
M3.03	Sketch common P&ID symbols for instrumentation as per ISA-5.1.	2	Applying

M3.04	Sketch P&ID for a boiler unit.	3	Applying
M3.05	Sketch P&ID for a chemical reactor	3	Applying
M3.06	Sketch P&ID for a heat exchanger	3	Applying
M3.07	Sketch P&ID for a distillation column	3	Applying
M3.08	Open ended experiment	3	Applying
CO4	Interpret piping plan and piping isometric drawings		
M4.01	Sketch the piping plan for the chemical reactor in M 3.05.	3	Applying
M4.02	Sketch the piping isometric for the chemical reactor in M 3.05.	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T ₁	Piping and Instrumentation Diagram Development; Moe Toghraei
R ₁	Dryden's Outlines Of Chemical Technology
R ₂	Perry's Chemical Engineers' Handbook; Robert H Perry

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/105/103105166/